

Competency

- To prove the theorems and properties of parallelograms and verify the other properties by inductive method.

Learning Outcomes

- To prove the theorems and properties of parallelograms
- To verify the other properties of parallelogram by experimental or inductive method.

Level-wise learning objectives

S.N.	LEVELS	OBJECTIVES
1.	Knowledge (K)	- To define parallelogram - To recall the types of parallelogram - To tell the properties of parallelogram - To recall the relations between corresponding parts of parallelogram (rectangle, square and rhombus).
2.	Understanding (U)	- To calculate the size of unknown angles based on properties of parallelograms
3.	Application (A)	- To prove the theorems on properties of parallelogram - To verify the properties of the parallelogram by inductive method
4.	High Ability (HA)	- To prove more theorems by drawing related diagrams/figure

Required Teaching Materials/ Resources

Colourful chart-paper, scale, scissors, pencil, geo-board, rubber bands, tangram, marker, models of measuring tape, ICT tools etc

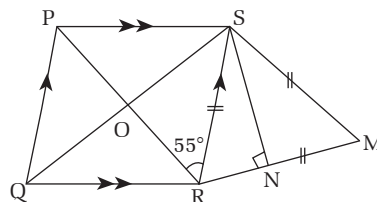
Pre-knowledge: Congruency of triangles, types of parallelograms, mid-point theorem etc.

Teaching Activities

1. Present the parallelograms on the geo-board by using rubber bands or make parallelograms using a rectangular piece of paper along with students and recall about parallelograms.
2. With the figure on the chart-paper, prove the theorems with students.
3. Rename the same theorem and give to students to prove them.
4. Select the students randomly and call to prove the theorem on the board.

Solution of selected problems from Vedanta Excel in Mathematics

1. In the given figure, PQRS is a rhombus and SRM is an equilateral triangle. If $SN \perp RM$ and $\angle PRS = 55^\circ$, find the size of $\angle QSN$.



Solution:

- (i) In rhombus PQRS, $\angle SOR = 90^\circ$ [Diagonals of rhombus bisect each other perpendicularly]

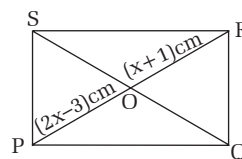
$$\therefore \angle OSR = 180^\circ - (90^\circ + 55^\circ) = 45^\circ$$

- (ii) In equilateral $\triangle SRN$; $\angle SRN = 60^\circ$ [Each angle of equilateral triangle]

$$\therefore \angle RSN = 180^\circ - (90^\circ + 60^\circ) = 30^\circ$$

$$\text{Hence, } \angle QSN = \angle OSR + \angle RSN = 45^\circ + 30^\circ = 75^\circ$$

2. In the adjoining figure, PQRS is a rectangle. If $OP = (2x - 3)$ cm and $OR = (x + 1)$ cm, find the length of diagonal QS.



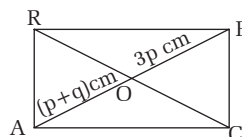
Solution:

- (i) $OP = OR$ [Diagonals of rectangle bisect each other]

$$\text{or, } 2x - 3 = x + 1 \quad \therefore x = 4 \text{ and } PR = OP + OR = (2x - 3) + (x + 1) = 10 \text{ cm}$$

- (ii) $PR = QS$ [Diagonals of rectangle are equal] $\therefore QS = 10 \text{ cm}$

3. RACE is a rectangle in which diagonal $RC = 18$ cm, $OA = (p + q)$ cm and $OE = 3p$ cm, find the values of p and q .



Solution:

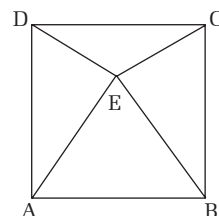
- (i) $AE = RC = 18 \text{ cm}$ [Diagonals of rectangle are equal]

- (ii) $OA = OE = 9 \text{ cm}$ [Diagonals of rectangle bisect each other]

$$\text{Now, } OE = 9 \text{ cm or, } 3p = 9 \text{ cm } \therefore p = 3 \text{ cm}$$

$$\text{Again, } OA = 9 \text{ cm or, } p + q = 9 \text{ cm or, } 3 \text{ cm} + q = 9 \text{ cm } \therefore q = 6 \text{ cm}$$

4. In the adjoining figure ABCD is a square and ABE is an equilateral triangle. Find the measure of $\angle ADE$ and $\angle DCE$.



Solution:

- (i) $\angle DAB = 90^\circ$ [Being an angle of the square]

- (ii) $\angle EAB = 60^\circ$ [Being an angle of the equilateral triangle]

$$\therefore \angle DAE = 90^\circ - 60^\circ = 30^\circ$$

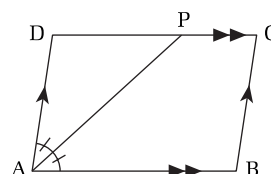
- (iii) $\angle ADE = \angle AED$ [$\because AD = AE$]

$$\text{Now, } \angle ADE + \angle AED + \angle DAE = 180^\circ$$

$$\text{or, } 2\angle ADE + 30^\circ = 180^\circ \therefore \angle ADE = 75^\circ$$

- (iv) $\angle BCE = 75^\circ$ [Same as the above process] $\therefore \angle DCE = 90^\circ - 75^\circ = 15^\circ$

5. In the given figure, ABCD is a parallelogram. AP bisects $\angle A$. Prove that $DP = BC$.



Solution:

Given: In the parallelogram ABCD; AP bisects $\angle A$.

To prove: $DP = BC$

Proof:

- (1) $\angle DAP = \angle BAP$ [AP bisects $\angle A$]
- (2) $\angle APD = \angle BAP$ [DC//AB, alternate angles]
- (3) $\angle DAP = \angle APD$ [From (1) and (2)]
- (4) $DP = AD$ [From (3), the base angles of $\triangle ADP$ are equal]
- (5) $AD = BC$ [Opposite sides of parallelogram are equal]
- (6) $DP = BC$ [From (4) and (5)]

6. **In the given figure, EXAM is a parallelogram, the bisector of $\angle A$ meets the mid-point of EM at P. Prove that $AX = 2AM$.**

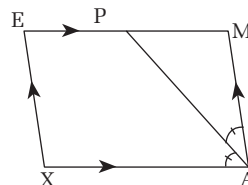
Solution:

Given: In the parallelogram EXAM; the bisector of $\angle A$ meets the mid-point of EM at P.

To prove: $AX = 2AM$

Proof:

- (1) $\angle XAP = \angle MAP$ [AP is the bisector of $\angle A$]
- (2) $\angle XAP = \angle MPA$ [EM//XA, alternate angles]
- (3) $\angle MAP = \angle MPA$ [From (1) and (2)]
- (4) $PM = AM$ [From (3), the base angles of $\triangle ADP$ are equal]
- (5) $ME = 2PM$ [P is the mid-point of ME]
- (6) $ME = AX$ [Opposite sides of parallelogram are equal]
- (7) $AX = 2AM$ [From (4),(5) and (6)]



7. **In the adjoining figure, ABCD is a rhombus in which CD is produced to E such that $CD = DE$. Prove that $\angle EAC = 90^\circ$.**

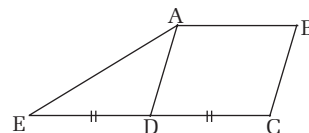
Solution:

Given: In the rhombus ABCD; CD is produced to E such that $CD = DE$.

To prove: $\angle EAC = 90^\circ$

Proof:

- (1) $AD = CD$ [Adjacent sides of the rhombus are equal]
- (2) $CD = DE$ [Given]
- (3) $AD = CD = DE$ [From (1) and (2)]
- (4) $\angle AEC = \angle EAD$ [From (3), $AD = DE$]
- (5) $\angle DAC = \angle ACE$ [From (3), $AD = CD$]
- (6) $\angle AEC + \angle EAC + \angle ACE = 180^\circ$ [Sum of angles of triangle]
- (7) $\angle EAD + \angle EAC + \angle DAE = 180^\circ$ [From (4), (5) and (6)]
- (8) $\angle EAC + \angle EAC = 180^\circ$ [From (7), $\angle EAD + \angle DAE = \angle EAC$]
- (9) $\angle EAC = 90^\circ$ [From (8)]



8. In the given figure, P and R are the mid-points of the sides AB and BC of $\triangle ABC$ respectively. If $PQ \parallel BC$, prove that $BP = RQ$.

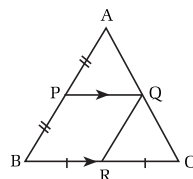
Solution:

Given: In $\triangle ABC$; P and R are the mid-points of the sides AB and BC respectively $PQ \parallel BC$.

To prove: $BP = RQ$

Proof:

- | | | |
|-----|---------------------------|---|
| (1) | $AQ = QC$ | [In $\triangle ABC$; $PQ \parallel BC$ and $AP = BP$] |
| (2) | $QR \parallel AB$ | [$AQ = QC$ and $BR = RC$] |
| (3) | $PQRB$ is a parallelogram | [From (2) and $PQ \parallel BC$] |
| (4) | $BP = RQ$ | [Opposite sides of the $\parallel$ gm $PQRB$] |



9. In the given quadrilateral PQRS, the mid-points of the sides PQ, QR, RS and SP are A, B, C and D respectively. Prove that ABCD is a parallelogram.

Solution:

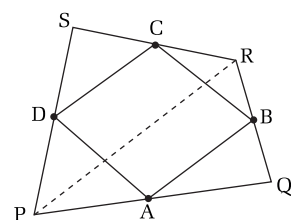
Given: In quadrilateral PQRS; the mid-points of sides PQ, QR, RS and SP are A, B, C and D respectively.

To prove: ABCD is a parallelogram

Construction: Diagonal PR of the quadrilateral PQRS is drawn

Proof:

- | | | |
|-----|--|--|
| (1) | In $\triangle PQR$, $AB \parallel PR$ and $AB = \frac{1}{2} PR$ | [AB joins the mid points of PQ and QR] |
| (2) | In $\triangle PSR$, $DC \parallel PR$ and $DC = \frac{1}{2} PR$ | [CD joins the mid points of RS and SP] |
| (3) | $AB \parallel DC$ and $AB = DC$ | [From (1) and (2)] |
| (4) | $BC \parallel AD$ and $BC = AD$ | [From (3)] |
| (5) | ABCD is a parallelogram | [From (3) and (4)] |



10. In the adjoining figure, P, Q, R and S are the mid-points of AB, BC, CD and AD respectively. Prove that PQRS is a parallelogram.

Solution:

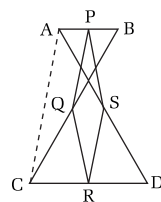
Given: P, Q, R and S are the mid-points of sides AB, BC, CD and AD respectively

To prove: PQRS is a parallelogram

Construction: AC is joined.

Proof:

- | | | |
|-----|--|--|
| (1) | In $\triangle ABC$, $PQ \parallel AC$ and $PQ = \frac{1}{2} AC$ | [PQ joins the mid points of AB and BC] |
| (2) | In $\triangle ADC$, $SR \parallel AC$ and $SR = \frac{1}{2} AC$ | [SR joins the mid points of AD and CD] |
| (3) | $PQ \parallel SR$ and $PQ = SR$ | [From (1) and (2)] |
| (4) | $QR \parallel PS$ and $QR = PS$ | [From (3)] |
| (5) | PQRS is a parallelogram | [From (3) and (4)] |



11. In the given figure, P, Q, R and S are the mid-points of AB, BC, CD and AD respectively. Prove that PQRS is a parallelogram.

Solution:

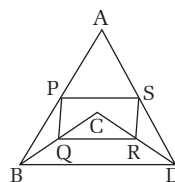
Given: P, Q, R and S are the mid-points of sides AB, BC, CD and AD respectively

To prove: PQRS is a parallelogram

Construction: BD is joined.

Proof:

- (1) In $\triangle ABD$, $PS \parallel BD$ and $PS = \frac{1}{2} BD$ [PS joins the mid points of AB and AD]
- (2) In $\triangle BCD$, $QR \parallel AC$ and $QR = \frac{1}{2} BC$ [QR joins the mid points of BC and CD]
- (3) $PS \parallel QR$ and $PS = QR$ [From (1) and (2)]
- (4) $PQ \parallel SR$ and $PQ = SR$ [From (3)]
- (5) PQRS is a parallelogram [From (3) and (4)]



12. In the given figure, PQRS is a square. A, B, C and D are the points on the sides PQ, QR, RS and SP respectively. If $AQ = BR = CS = DP$, prove that ABCD is also a square.

Solution:

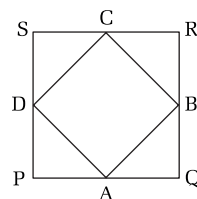
Given: PQRS is a square. $AQ = BR = CS = DP$

To prove: ABCD is the square.

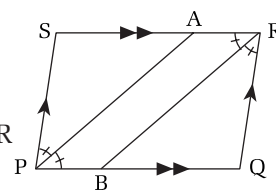
Proof:

- (1) $AQ = BR = CS = DP$ [Given]
- (2) $PA = BQ = CR = SD$ [Remaining parts of equal sides]
- (3) In $\triangle AQB$ and $\triangle BRC$ [From (1)]
 - (i) $\angle AQB = \angle BRC$ [Both are right angles]
 - (ii) $\angle AQB = \angle BRC$ [From (2)]
 - (iii) $BQ = CR$
- (4) $\triangle AQB \cong \triangle BRC$ [By S.A.S axiom]
- (5) $AB = BC$ and $\angle ABQ = \angle BCR$ [Corresponding parts of congruent triangles]
- (6) $\angle CBR + \angle BCR = 90^\circ$ [Sum of acute angles in rt. $\angle$ ed $\triangle BCR$]
- (7) $\angle CBR + \angle ABQ = 90^\circ$ [From (5) and (6)]
- (8) $\angle CBR + \angle ABQ + \angle ABC = 180^\circ$ [Being the parts of a straight angle]

$\therefore \angle ABC = 90^\circ$
- (9) $\triangle BCR \cong \triangle CDS$, $\triangle CDS \cong \triangle AQB$ [By S.A.S axiom]
- (10) $BC = CD$ and $CD = AD$ [Corresponding sides of congruent triangles]
- (11) ABCD is a square [From (5), (9) and (10)]



13. In the given parallelogram PQRS, PA bisects $\angle P$ and RB bisects $\angle R$. Prove that $PA \parallel BR$.



Solution:

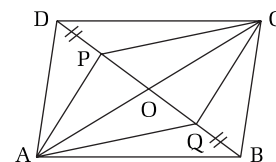
Given: In the parallelogram PQRS, PA bisects $\angle A$ and RB bisects $\angle R$

To prove: $PA \parallel BR$

Proof:

- (1) $\angle SPQ = 2\angle APQ$ and $\angle QRS = 2\angle BRS$ [Given]
- (2) $\angle SPQ = \angle QRS$ [Opposite angles of parallelogram]
- (3) $\angle APQ = \angle BRS$ [From (1) and (2)]
- (4) $\angle APQ = \angle SAP$ [SR//PQ, alternate angles]
- (5) $\angle BRS = \angle SAB$ [From (3) and (4)]
- (6) $PA \parallel BR$ [Corresponding angles are equal]

14. ABCD is a parallelogram. P and Q are two points on the diagonal BD such that $DP = QB$. Prove that APCQ is a parallelogram.



Solution:

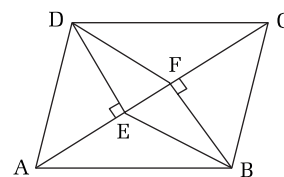
Given: In the parallelogram ABCD, P and Q are two points on the diagonals BD such that $DP = QB$

To prove: APCQ is a parallelogram

Proof:

- (1) $OA = OC$ and $OD = OB$ [Diagonals of parallelogram bisect each other]
- (2) $DP = QB$ [Given]
- (3) $OA = OB$ and $OP = OQ$ [$OD - DP = OP$, $OB - QB = OQ$]
- (4) APCQ is a parallelogram [From (3), diagonals are bisected each other]

15. ABCD is a parallelogram. $DE \perp AC$ and $BF \perp AC$. Prove that BEDF is a parallelogram.



Solution:

Given: ABCD is a parallelogram. $DE \perp AC$ and $BF \perp AC$

To prove: BEDF is a parallelogram

Proof:

- (1) In $\triangle DAE$ and $\triangle BCF$
 - (i) $\angle AED = \angle BFC$ [Both are right angles]
 - (ii) $\angle DAE = \angle BCF$ [DA//BC, alternate angles]
 - (iii) $AD = BC$ [Opposite sides of parallelogram]
- (2) $\triangle DAE \cong \triangle BCF$ [By A.A.S. axiom]
- (3) $AP = QC$ [Corresponding sides of congruent triangles]
- (4) $DE \parallel FB$ [$\angle AED = \angle BFC$, alternate exterior angles are equal]
- (5) $DF = EB$, $DF \parallel EB$ [From (3) and (4)]
- (6) BEDF is a parallelogram [From (3), (4) and (5)]

16. In parallelogram PQRS, the bisectors of $\angle PQR$ and $\angle PSR$ meet the diagonal PR at M and N respectively. Prove that MQNS is a parallelogram.

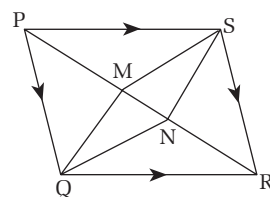
Solution:

Given: In //gm PQRS, the bisectors of $\angle PQR$ and $\angle PSR$ meet the diagonal PR at M and N respectively i.e., $\angle PQM = \angle RSN$ and $\angle PSN = \angle RSN$

To prove: MQNS is a parallelogram

Proof:

- (1) In $\triangle PQM$ and $\triangle RSN$
 - (i) $\angle PQM = \angle RSN$ [Given and $\angle PQR = \angle PSR$]
 - (ii) $PQ = RS$ [Opposite sides of parallelogram]
 - (iii) $\angle QPM = \angle SRN$ [PQ//SR, alternate angles]
- (2) $\triangle PQM \cong \triangle RSN$ [By A.S.A. axiom]
- (3) $MQ = SN$, $\angle PMQ = \angle SNR$ [Corresponding parts of congruent triangles]
- (4) $MQ \parallel SN$ [From (3), alternate exterior angles are equal]
- (5) $MS = QN$, $MS \parallel QN$ [From (3) and (4)]
- (6) MQNS is a parallelogram [From (3), (4) and (5)]



17. In the given figure, ABCD is a parallelogram. If P and Q are the points of trisection of diagonal BD, prove that PAQC is a parallelogram.

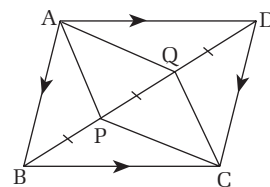
Solution:

Given: In parallelogram ABCD, P and Q are the points of trisections of the diagonal BD. i.e., $BP = PQ = QD$

To prove: PAQC is a parallelogram

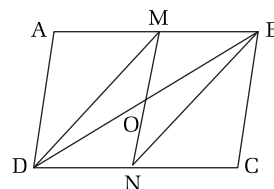
Proof:

- (1) In $\triangle ABP$ and $\triangle CDQ$
 - (i) $AB = CD$ [Opposite sides of parallelogram]
 - (ii) $\angle ABP = \angle CDQ$ [AB//CD, alternate angles]
 - (iii) $BP = QD$ [Given]
- (2) $\triangle ABP \cong \triangle CDQ$ [By S.A.S. axiom]
- (3) $AP = CQ$, $\angle APB = \angle CQD$ [Corresponding parts of congruent triangles]
- (4) $AP \parallel QC$ [From (3), alternate exterior angles are equal]
- (5) $AQ = PC$, $AQ \parallel PC$ [From (3) and (4)]
- (6) PAQC is a parallelogram [From (3), (4) and (5)]



18. In the figure, ABCD is a parallelogram. M and N are the mid-points of the sides AB and DC respectively. Prove that

- (i) MBCN is a parallelogram
- (ii) DMBN is a parallelogram
- (iii) DB and MN bisect each other at O.



Solution:

Given: In parallelogram ABCD, M and N are the mid-points of the sides AB and DC respectively

To prove: (i) MBCN is a parallelogram

(ii) DMBN is a parallelogram

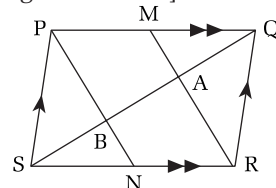
(iii) DB and MN bisect each other

Proof:

- (1) $AB = DC$ and $AB \parallel DC$ [Opposite sides of parallelogram]
- (2) $MB = CN$ and $MB \parallel CN$ [M and N are the mid-points of AB and DC]
- (3) $MN = BC$ and $MN \parallel BC$ [From (2)]
- (4) MBCN is a parallelogram [From (2) and (3)]
- (5) $DM = BN$ and $DM \parallel BN$ [$MB = DN$ and $MB \parallel DN$]
- (6) DMBN is a parallelogram [From (5)]
- (7) DB and MN bisect each other [Being the diagonals of parallelogram DMBN]

19. In the given parallelogram PQRS, M and N are the mid-points of the sides PQ and SR respectively. Prove that

(i) PNRM is a parallelogram (ii) $QA = AB = BS$

**Solution:**

Given: In parallelogram ABCD, M and N are the mid-points of the sides PQ and SR respectively

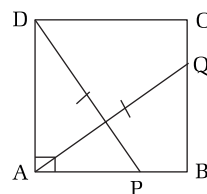
To prove: (i) PNRM is a parallelogram

(ii) $QA = AB = BS$

Proof:

- (1) $PM = NR$ and $PM \parallel NR$ [$PQ = SR$, $PQ \parallel SR$ and given]
- (2) $PN = MR$ and $PN \parallel MR$ [From (1)]
- (3) PNRM is a parallelogram [From (1) and (2)]
- (4) $QA = AB$ [In $\triangle PQB$, $PM = QM$ and $MA \parallel PB$]
- (5) $AM = BS$ [In $\triangle SAR$, $SN = RN$ and $BN \parallel AR$]
- (6) $QA = AB = BS$ [From (4) and (5)]

20. ABCD is a square. P and Q are any points on the sides AB and BC respectively. If $AQ = DP$, prove that AQ and DP are perpendicular to each other.

**Solution:**

Given: In square ABCD, P and Q are the points on the sides AB and BC respectively

To prove: AQ and DP are perpendicular to each other. $AQ = DP$

Proof:

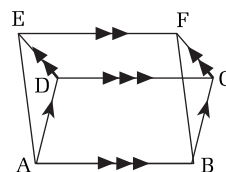
- (1) In $\triangle DAP$ and $\triangle ABQ$
 - (i) $\angle DAP = \angle ABQ$ (R) [Both are right angles]
 - (ii) $DP = AQ$ (H) [Given]
 - (iii) $DA = AB$ (S) [Adjacent sides of the square ABCD]
- (2) $\triangle DAP \cong \triangle ABQ$ [By R.H.S. axiom]
- (3) $\angle ADP = \angle BAQ$ [Corresponding angles of congruent triangles]
- (4) $\angle DAQ + \angle ADP = 90^\circ$ [From (3) and $\angle DAP = \angle DAQ + \angle QAP = 90^\circ$]
- (5) $\angle AOD = 90^\circ$ [From (4), remaining angle of $\triangle AOD$]

Hence, AQ and DP are perpendicular each other.

21. In the figure along side, ABCD and DEFC are parallelograms.

Prove that

(i) $AE = BF$ (ii) $\triangle ADE \cong \triangle BCF$



Solution:

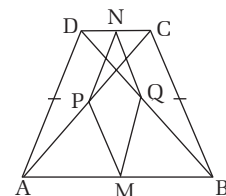
Given: ABCD and DEFC are parallelograms.

To prove: (i) $AE = BF$ (ii) $\triangle ADE \cong \triangle BCF$

Proof:

- | | | |
|-------|--|--|
| (1) | $AB = DC$, $AB \parallel DC$ | [Opposite sides of parallelogram ABCD] |
| (2) | $DC = EF$, $DC \parallel EF$ | [Opposite sides of parallelogram DEFC] |
| (3) | $AB = EF$ and $AB \parallel EF$ | [From (1) and (2)] |
| (4) | $AE = BF$ and $AE \parallel BF$ | [From (3)] |
| (6) | In $\triangle ADE$ and $\triangle BCF$ | |
| (i) | $AE = BF$ (S) | [From (4)] |
| (ii) | $AD = BC$ (S) | [Opposite sides of parallelogram ABCD] |
| (iii) | $DE = CF$ (S) | [Opposite sides of parallelogram DEFC] |
| (7) | $\triangle ADE \cong \triangle BCF$ | [By S.S.S. axiom] |

22. In the given quadrilateral ABCD, $AD = BC$, P and Q are the mid-points of the diagonals AC and BD respectively. If M and N are the mid-points of the sides DC and AB respectively, prove that PMQN is a parallelogram.



Solution:

Given: In quadrilateral ABCD, $AD = BC$, P and Q are the mid-points of the diagonals AC and BD respectively, M and N are the mid-points of the sides DC and AB respectively.

To prove: PMQN is a rhombus

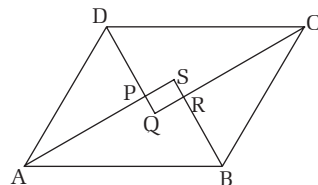
Proof:

$NP \parallel DA$ and $NP = \frac{1}{2} DA$ [Applying mid-point theorem in $\triangle CDA$]

$QM \parallel DA$ and $QM = \frac{1}{2} DA$ [Applying mid-point theorem in $\triangle BDA$]

- | | | |
|-----|---------------------------------|--------------------|
| (1) | $NP \parallel QM$ and $NP = QM$ | [From (1) and (2)] |
| (2) | $PM \parallel NQ$ and $PM = NQ$ | [From (3)] |
| (3) | $NP = PM = QM = NQ$ | [$AD = BC$] |
| (4) | PMQN is a rhombus | [From (5)] |

23. In the adjoining figure, ABCD is a parallelogram. AS, BS, CQ and DQ are the bisectors of $\angle A$, $\angle B$, $\angle C$ and $\angle D$ respectively. Prove that PQRS is a rectangle.



Solution:

Given: In parallelogram ABCD; AS, BS, CQ and DQ are the bisectors of $\angle A$, $\angle B$, $\angle C$ and $\angle D$ respectively.

To prove: PQRS is a rectangle

Proof:

- (1) $\angle DAB = 2\angle DAS = 2\angle BAS$, $\angle ABC = 2\angle ABS = 2\angle CBS$, $\angle BCD = 2\angle BCQ = 2\angle DCQ$
and $\angle ADC = 2\angle ADQ = 2\angle CDQ$ [Given]
- (2) $\angle DAB + \angle ABC = 180^\circ$ [AD//BC, co-interior angles]
- (3) $2\angle BAS + 2\angle ABS = 180^\circ$ [From (1) and (2)]
 $\therefore \angle BAS + \angle ABS = 90^\circ$
- (4) $\angle BAS + \angle ABS + \angle ASB = 180^\circ$ [Sum of angles of triangle ABS]
- (5) $\angle ASB = 90^\circ$ [From (3) and (4)]
- (6) $\angle CQD = \angle SPQ = \angle SRQ = 90^\circ$ [Same as above process]
- (7) PQRS is a rectangle [From (5) and (6)]

24. Prove that the diagonals of a rectangle are equal.

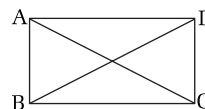
Solution:

Given: ABCD is a rectangle in which AC and BD are the diagonals

To prove: $AC = BD$

Proof:

- (1) In $\triangle ABC$ and $\triangle DCB$
 - (i) $AB = DC$ (S) [Opposite sides of rectangle are equal]
 - (ii) $\angle ABC = \angle DCB$ (A) [Both are right angles]
 - (iii) $BC = BC$ (S) [Common side]
- (2) $\triangle ABC \cong \triangle DCB$ [By S.A.S. axiom]
- (3) $AC = BD$ [Corresponding sides of congruent triangles]



25. If the diagonals of a rhombus are equal, prove that it is a square.

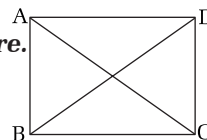
Solution:

Given: In rhombus ABCD, $AC = BD$

To prove: ABCD is a square

Proof:

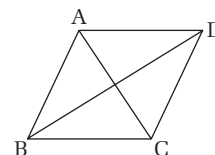
- (1) In $\triangle ABC$ and $\triangle DCB$
 - (i) $AB = DC$ (S) [Opposite sides of rectangle are equal]
 - (ii) $BC = BC$ (S) [Common side]
 - (iii) $AC = BD$ (S) [Given]
- (2) $\triangle ABC \cong \triangle DCB$ [By S.S.S. axiom]
- (3) $\angle ABC = \angle DCB$ [Corresponding angles of congruent triangles]
- (4) $\angle ABC + \angle DCB = 180^\circ$ [AB//DC and co-interior angles]
- (5) $\angle ABC = 90^\circ$ [From (3) and (4)]
- (6) ABCD is a square [From (5) and $AB = BC = CD = DA$]



26. Prove that the diagonals of a rhombus bisect each other perpendicularly.

Solution:

Given: In rhombus ABCD, the diagonals AC and BD intersect at O.



To prove: Diagonals AC and BD bisect each other perpendicularly i.e., $OA = OC$, $OB = OD$
and $\angle AOB = 90^\circ$

Proof:

- (1) In $\triangle AOB$ and $\triangle AOD$
- (i) $AB = AD$ (S) [Sides of rhombus are equal]
- (ii) $\angle BAO = \angle DAO$ (A) [Diagonal AC bisects $\angle A$]
- (iii) $OA = OA$ (S) [Common side]
- (2) $\triangle AOB \cong \triangle AOD$ [By S.A.S. axiom]
- (3) $OB = OD$ [Corresponding sides of congruent triangles]
- (4) $\angle AOB = \angle AOD$ [Corresponding angles of congruent triangles]
- (5) $\angle AOB = \angle AOD = 90^\circ$ [Being linear pair]
- (6) $\triangle AOB \cong \triangle BOC$ [Same as above process]
- (7) $OA = OC$ [Corresponding sides of congruent triangles]
- (8) AC and BD bisect each other perpendicularly at O [From (3), (5) and (7)]

27. In the adjoining quadrilateral PQRS, $\angle P = \angle R$ and $\angle Q = \angle S$. Show that PQRS is a parallelogram.

Solution:

Given: In quadrilateral PQRS, $\angle P = \angle R$ and $\angle Q = \angle S$

To prove: PQRS is a parallelogram

Proof:

- (1) $\angle P = \angle R$ and $\angle Q = \angle S$ [Given]
- (2) $\angle P + \angle R + \angle Q + \angle S = 360^\circ$ [Sum of angles of quadrilateral]
- (3) $\angle R + \angle R + \angle S + \angle S = 360^\circ$ [From (i) and (ii)]
 $\therefore \angle R + \angle S = 180^\circ$
- (4) $PS \parallel QR$ [From (3) co-interior angles are supplementary]
- (5) $PQ \parallel SR$ [Same as above process]
- (6) PQRS is a parallelogram [From (4) and (5)]

28. In the given quadrilateral ABCD, $AO = OC$ and $BO = OD$. Prove that ABCD is a parallelogram.

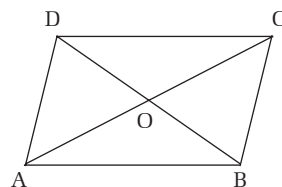
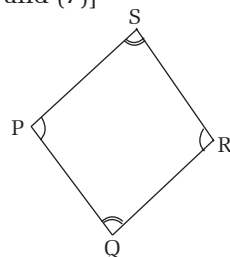
Solution:

Given: In quadrilateral PQRS, $OA = OC$ and $OB = OD$

To prove: ABCD is a parallelogram

Proof:

- (1) In $\triangle AOD$ and $\triangle BOC$
- (i) $OA = OC$ (S) [Given]
- (ii) $\angle AOD = \angle BOC$ (A) [Vertically opposite angles are equal]
- (iii) $OD = OB$ (S) [Given]
- (2) $\triangle AOD \cong \triangle BOC$ [By S.A.S. axiom]
- (3) $AD = BC$ and $\angle ADO = \angle OBC$ [Corresponding parts of congruent triangles]
- (4) $AD \parallel BC$ [From (3), alternate angles are equal]
- (5) $AB \parallel DC$ and $AB = DC$ [From (3) and (4)]
- (6) ABCD is a parallelogram [From (3), (4) and (5)]



29. In the adjoining parallelogram $ABCD$, P and Q are the mid-points of the sides AD and BC respectively. Prove that BP and QD trisect the diagonal AC at X and Y respectively.

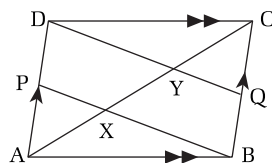
Solution:

Given: In parallelogram $ABCD$, P and Q are the mid-points of the sides AD and BC respectively

To prove: BP and QD trisect AC at X and Y i.e., $AX = XY = YC$

Proof:

- (1) $PD = BQ$ and $PD \parallel BQ$ [$AD = BC$, $AD \parallel BC$ and given]
- (2) $PB = DQ$ and $B \parallel DQ$ [From (1)]
- (3) $AX = XY$ [In $\triangle ADY$, $AP = PD$ and $PX \parallel DY$]
- (4) $XY = YC$ [In $\triangle BCX$, $BQ = QC$ and $YQ \parallel XB$]
- (5) $AX = XY = YC$ [From (3) and (4)]



30. In the given parallelogram $PQRS$, M and N are the mid-points of the sides SR and QR respectively. If the diagonal PR and QS intersect at H , prove that $MRNH$ is a parallelogram.

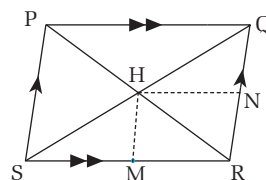
Solution:

Given: In parallelogram $PQRS$, M and N are the mid-points of the sides SR and QR respectively

To prove: $MRNH$ is a parallelogram

Proof:

- (1) $HP = HR$ and $HQ = HS$ [Diagonals of parallelogram bisect each other]
- (2) $PS \parallel QR$ and $PQ \parallel SR$ [Opposite sides of parallelogram $PQRS$]
- (3) In $\triangle QSR$, $HN \parallel SR$ [HN joins mid-points of the sides QS and QR]
- (4) In $\triangle PSR$, $HM \parallel PS$ [HM joins mid-points of the sides PR and SR]
- (5) $HN \parallel SR$ and $HM \parallel QR$ [From (2), (3) and (4)]
- (6) $MRNH$ is a parallelogram [From (5)]



31. In the figure alongside, $ABCD$ is a quadrilateral in which $OA = OC$, $OB = OD$ and $\angle AOB = \angle AOD = \angle DOC = \angle COB = 90^\circ$. Prove that $ABCD$ is a rhombus.

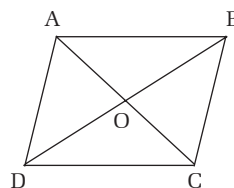
Solution:

Given: In quadrilateral $ABCD$, $OA = OC$, $OB = OD$ and $\angle AOB = \angle AOD = \angle DOC = \angle COB = 90^\circ$

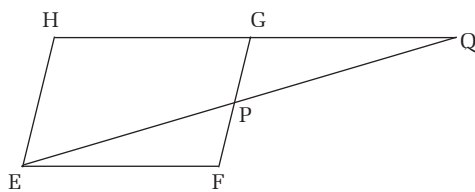
To prove: $ABCD$ is a rhombus

Proof:

- (1) In $\triangle AOB$ and $\triangle BOC$
 - (i) $OA = OC$ (S) [Given]
 - (ii) $\angle AOB = \angle BOC$ (A) [Both are right angles]
 - (iii) $OB = OB$ (S) [Common side]
- (2) $\triangle AOB \cong \triangle BOC$ [By S.A.S. axiom]
- (3) $AB = BC$ [Corresponding sides of congruent triangles]
- (4) $\triangle BOC \cong \triangle COD$ and $\triangle COD \cong \triangle AOD$ [Same as above process]
- (5) $BC = CD = AD$ [Corresponding sides of congruent triangles]
- (6) $ABCD$ is a rhombus [From (3) and (5)]



32. In the adjoining figure, EFGH is a parallelogram and P is the mid-point of FG. EP and HG are produced to meet at Q. Prove that $AQ = 2HG$.



Solution:

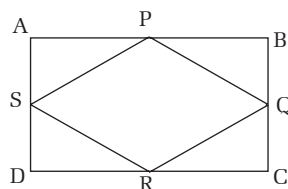
Given: In parallelogram EFGH, P is the mid-point of FG. EP and HG produced meet at Q.

To prove: $HQ = 2HG$

Proof:

- | | |
|--|--|
| (1) In $\triangle EFP$ and $\triangle GPQ$ | |
| (i) $\angle PEF = \angle PQG$ (A) | [HQ//EF, alternate angles] |
| (ii) $\angle EPF = \angle GPQ$ (A) | [Vertically opposite angles are equal] |
| (iii) $FP = GP$ (S) | [Given] |
| (2) $\triangle EFP \cong \triangle GPQ$ | [By A.A.S. axiom] |
| (3) $EF = GQ$ | [Corresponding sides of congruent triangles] |
| (4) $EF = HG$ | [Opposite sides of parallelogram EFGH] |
| (5) $HQ = 2HG$ | [From (3) and (4)] |

33. In the adjoining figure, ABCD is a rectangle and P, Q, R and S are the mid-points of AB, BC, CD and DA respectively. Prove that PQRS is a rhombus.



Solution:

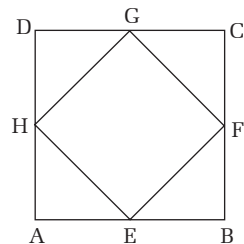
Given: In rectangle ABCD; P, Q, R and S are the mid-points of AB, BC, CD and DA respectively

To prove: PQRS is a rhombus

Proof:

- | | |
|---|--|
| (1) $SA = SD = BQ = QC$ | [S and Q are mid-points of AD and BC] |
| (2) $DR = CR = AP = BP$ | [R and P are mid-points of DC and AB] |
| (3) In $\triangle SDR$ and $\triangle RCQ$ | |
| (i) $SD = QC$ (S) | [From (1)] |
| (ii) $\angle SDR = \angle RCQ$ (A) | [Both are right angles] |
| (iii) $DR = CR$ (S) | [From (2)] |
| (4) $\triangle SDR \cong \triangle RCQ$ | [By S.A.S. axiom] |
| (5) $SR = QR$ | [Corresponding sides of congruent triangles] |
| (6) $\triangle RCQ \cong \triangle PQB \cong \triangle ASP$ | [Same as above process] |
| (7) $QR = PQ = PS$ | [Corresponding sides of congruent triangles] |
| (8) PQRS is a rhombus | [From (5) and (7)] |

34. In the given square ABCD, E, F, G and H are the mid-points of AB, BC, CD and DA respectively. Prove that EFGH is also a square.



Solution:

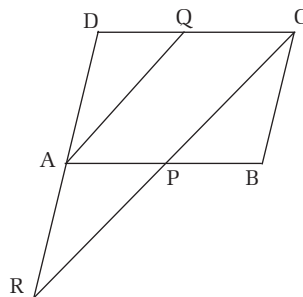
Given: In square ABCD; E, F, G and H are the mid-points of AB, BC, CD and DA respectively

To prove: EFGH is a square

Proof:

- | | |
|--|---|
| (1) In $\triangle HAE$ and $\triangle EBF$ | |
| (i) $HA = BF$ (S) | [H and F are mid-points of AD and BC, $AD = BC$] |
| (ii) $\angle HAE = \angle EBF$ (A) | [Both are right angles] |

- (iii) $AE = BE$ (S) [E is the mid-point of AB]
 (2) $\triangle HAE \cong \triangle EBF$ [By S.A.S. axiom]
 (3) $HE = EF$ and $\angle AEH = \angle BEF$ [Corresponding parts of congruent triangles]
 (4) $\triangle EBF \cong \triangle FCG \cong \triangle HDG$ [Same as above process]
 (5) $EF = FG = HG$ [Corresponding sides of congruent triangles]
 (6) $\angle AEH = \angle BEF = 45^\circ$ [Acute angles of isosceles right angled triangles]
 (7) $\angle AEH + \angle BEF + \angle HEF = 180^\circ$ [Being parts of straight angle]
 (8) $\angle HEF = 90^\circ$ [From (6) and (7)]
 (9) PQRS is a square [From (3), (5) and (8)]



35. In the given figure, ABCD is a parallelogram. If P and Q are the mid-points of the sides AB and DC respectively, prove that $RC = 2AQ$.

Solution:

Given: In parallelogram ABCD; P and Q are the mid-points of AB and DC respectively

To prove: $RC = 2AQ$

Proof:

- (1) $QC = AP$ and $QC \parallel AP$ [DC = AB, DC $\parallel$ AB and given]
 (2) $AQ = PC$ and $AQ \parallel PC$ [From (1)]
 (3) $DA = AR$ [DQ = QC and AQ $\parallel$ PC]
 In $\triangle DRC$, $AQ = \frac{1}{2}RC$ [AQ joins the mid-points of DR and DC]
 $\therefore RC = 2AQ$

36. In the given figure, P, Q, R and S are the mid-points of AB, BC, CD and AD respectively. Prove that PQRS is a parallelogram.

Solution:

Given: $AB = AC$, $PB = CM$ and $PQ \parallel AM$

To prove: PM and QC bisect each other. i.e., $OP = OM$ and $OQ = OC$

Proof:

- (1) $\angle ABC = \angle ACB$ [AB = AC]
 (2) $\angle PQB = \angle ACB$ [PQ $\parallel$ AC, corresponding angles]
 (3) $\angle ABC = \angle PQB$ [From (1) and (2)]
 (4) $PB = PQ$ [From (3)]
 (5) $PB = CM$ [Given]
 (6) $PQ = CM$ [From (4) and (5)]
 (7) In $\triangle POQ$ and $\triangle COM$
 (i) $\angle POQ = \angle COM$ (A) [Vertically opposite angles are equal]
 (ii) $\angle OPQ = \angle OMC$ (A) [PQ $\parallel$ CM, alternate angles]
 (iii) $PQ = CM$ (S) [From (6)]
 (8) $\triangle POQ \cong \triangle COM$ [By A.A.S. axiom]
 (9) $OP = OM$ and $OQ = OC$ [Corresponding sides of congruent triangles]

Hence, PM and QC bisect each other at O.

